

Herron Hà-Vi Nguyen

herron@stanford.edu · github.com/herronoui · [LinkedIn](#) · Stanford, CA · Available June to September 2026

SUMMARY

B.S. Mechanical Engineering candidate at Stanford (Expected Winter 2027), applying to the AA co-terminal M.S. with a focus on safe autonomous systems. Research involves safety-critical human-robot interaction (CHARM Lab, Magic Chair) and probabilistic state estimation for autonomous decision-making (AA 273 Bayesian MOT + chance-constrained overtaking). Independently implemented a Singer UKF, multi-object tracking pipeline, and Monte Carlo risk-estimation framework from scratch in one quarter on F1 racing telemetry data. Currently training for a private pilot license, bringing firsthand operational context in safety-critical aviation to research on autonomous aircraft systems, including runway-environment perception and failure-mode awareness. Strong Python, PyTorch, and MATLAB skills; fast, self-directed learner comfortable at the cross-section of controls, deep learning, and real-world autonomous systems.

EDUCATION

Stanford University

Expected Winter 2027 | Stanford, CA

B.S., Mechanical Engineering · B.A., French · **Silver Honor**, Intl. Astronomy & Astrophysics Competition 2025 · Student Pilot in Training

Coursework: State Estimation for Robotics & Perception, Control Systems, Principles of Safety-Critical Systems (audit), Haptic Systems Design and Control, Mechatronics, Advanced Feedback Control Design, Dynamics, Circuits & Electronics

TECHNICAL SKILLS

Programming & ML: Python (NumPy, SciPy, Matplotlib, PyTorch, pytest), MATLAB/Simulink, C++ , data pipelines, test automation, statistical analysis

Autonomy & Controls: Bayesian state estimation, Kalman filtering (UKF/EKF/Singer), multi-object tracking (Hungarian, Mahalanobis gating), chance-constrained optimization, Monte Carlo risk estimation, reachability analysis, control barrier functions, sensor fusion

Hardware & Systems: Pneumatic system characterization, embedded firmware control loops, motor and actuator systems, hardware-in-the-loop testing, mechatronics integration, PCB-level debugging, signal integrity analysis

Design & Fabrication: SolidWorks, Fusion 360, FEA (static, buckling, Von Mises), GD&T, DFM/DFA, rapid prototyping, 3D printing (CF-PLA, FDM, multimaterial)

EXPERIENCE

Stanford CHARM Lab, Dept. of Mechanical Engineering *ME Research Rotator*

Jan 2026 to Present | Stanford, CA

- Leads hardware design and validation of a safety-critical soft robotic patient-transfer system (Magic Chair), where system failure directly impacts patient safety, motivating rigorous, deployment-ready engineering with formal safety considerations integrated across mechanical, electrical, and software subsystems.
- Designed and executed a controlled pneumatic characterization experiment isolating the soft everting structure to measure intrinsic pressure-to-growth behavior with and without chair attachment; results directly inform minimum pressure thresholds and closed-loop actuation control setpoints for safe patient lifting.
- Integrates sensor systems, embedded firmware control loops, motor drivers, and mechatronics subsystems; performs iterative design verification and hardware qualification; led first system deployment at the Stanford Robotics Center in collaboration with MIT ME.

TBrain AI *QC Lead Engineer & Project Manager*

Jun 2025 to Present | San Francisco, CA

- Built and owned a structured design verification process for 550+ GD&T-compliant production drawings; applied statistical process control to sustain **96–99% accuracy** and cut defect rate by **40%**.
- Eliminated **98%** of pre-release manufacturing risks through systematic annotation and tolerance review; led 5-person international team to deliver a 10,000+ unit follow-on contract.

Institut Supérieur d'Électronique de Paris *ME Research Intern*

Sep 2024 to Jan 2025 | Paris, France

- Independently learned GaN semiconductor physics from first principles and built a reduced-order performance model for MEMS vibration energy harvesters; implemented frequency response analysis and characterized nonlinear power trade-offs across variable conditions in MATLAB, producing full design documentation per iteration.

SELECTED PROJECTS

Bayesian Multi-Object Tracking & Risk-Aware Autonomy, F1 Telemetry *State Estimation for Robotics & Perception, Stanford University* [\[code\]](#)

- Independently implemented a Singer UKF, Mahalanobis-gated Hungarian MOT pipeline, and Monte Carlo chance-constraint decision-making framework from scratch in one quarter on 87 real F1 events (25 races, Charles Leclerc as ego agent).
- Formulated autonomous overtaking as a probabilistic safety problem: ego agent overtakes only when estimated collision probability falls below a calibrated risk threshold, a closed-loop, system-level safety paradigm directly analogous to the run-time failure monitor and fallback controller structure in Neural Reachable Tubes research.
- Achieved **80% false-safe reduction**; validated covariance calibration at 71%/96% consistency across Monte Carlo trials; pipeline generalizes to multi-agent autonomous decision-making under uncertainty.

Generative Design Autonomous Delivery Robot *Advanced Additive Manufacturing Design, Stanford University*

- Designed and fabricated a fully autonomous mobile robot with a generatively-optimized load-bearing ABS chassis: modeled six real-world dynamic load cases in FEA; implemented embedded motor control firmware and full mechatronics wiring; assembled and validated complete system under live payload conditions.

Motor-Driven Leadscrew Lifting Machine *Mechanical Systems Design, Stanford University*

- Designed a fully integrated electromechanical system from first principles: motor setpoint selection, gear train design, and system-level post-build validation achieving $F = 275\text{ N}$ at constant 6 mm/s ; first-principles failure analysis across bevel gears, motor shaft bending, and leadscrew buckling at $P_{cr} \approx 13,810\text{ N}$; validated via MATLAB performance modeling against analytical model.